

WINDSOR, ON, Canada

WMO: 715380

Lat: 42.2788N

Lon: 82.9494W

Elev: 622

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.5	7.6	-7.5	3.8	3.4	-1.9	5.1	9.5	31.2	26.1	27.5	25.2	11.0	240	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	89.5	73.7	86.8	72.4	84.3	71.0	76.4	85.5	74.7	83.1	73.1	80.9	12.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.7	128.2	81.7	72.0	121.2	79.6	70.5	114.8	78.2	40.3	85.6	38.6	83.5	37.1	80.9	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.5	22.6	20.2	DB	-3.0	94.2	7.2	3.0	-8.2	96.4	-12.4	98.2	-16.4	99.9	-21.6	102.1	(4)
(5)			WB	-4.0	79.8	6.9	1.8	-8.9	81.1	-13.0	82.2	-16.9	83.2	-21.9	84.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	25.4	27.7	36.2	48.0	59.7	69.7	73.7	72.0	65.3	53.5	41.4	31.3		
	DBStd	18.99	10.82	10.15	10.28	8.71	8.32	6.81	5.37	5.04	6.92	8.22	8.65	9.07		
	HDD50	2901	762	626	445	136	12	0	0	0	1	55	283	581		
	HDD65	6128	1226	1045	893	513	211	30	2	6	81	369	709	1043		
	CDD50	3064	1	1	18	77	311	592	735	681	459	163	24	2		
	CDD65	819	0	0	1	3	46	173	273	222	89	12	0	0		
	CDH74	6152	0	0	7	46	399	1374	2209	1488	559	70	0	0		
	CDH80	1792	0	0	0	6	105	445	708	380	137	11	0	0		
(14) Wind	WSAvg	9.7	11.7	11.3	11.1	11.3	9.7	8.7	7.8	7.1	7.7	9.4	10.6	10.8		
(15) Precipitation	PrecAvg	36.2	2.1	2.2	3.0	3.2	3.1	4.0	3.5	3.5	3.3	2.3	3.0	3.2		
	PrecMax	45.4	4.4	6.1	5.3	6.4	5.7	6.6	9.8	8.0	9.9	4.3	6.2	6.4		
	PrecMin	25.6	0.7	0.3	1.1	1.3	1.2	1.1	1.5	0.6	0.7	0.6	0.9	0.8		
	PrecStd	5.2	1.0	1.4	1.1	1.2	1.1	1.4	1.9	1.9	1.9	1.1	1.5	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.8	59.3	72.1	80.0	87.2	92.1	93.5	91.1	88.7	80.9	66.4	58.2		
		MCWB	52.4	52.4	61.6	65.1	69.7	73.9	76.4	74.5	73.3	68.2	57.2	55.4		
	2%	DB	50.2	50.6	63.0	72.9	82.9	88.6	89.8	87.0	84.2	75.0	61.3	52.6		
		MCWB	48.7	46.3	55.1	59.5	68.0	72.2	74.0	73.8	70.6	64.8	55.5	49.7		
	5%	DB	44.1	45.0	57.1	67.9	78.7	85.3	86.8	84.2	80.3	70.5	58.1	47.4		
		MCWB	41.4	41.0	50.4	57.1	66.4	70.7	72.9	72.0	68.4	61.9	53.0	44.2		
	10%	DB	39.1	40.1	51.3	62.6	73.9	81.7	84.0	81.5	76.8	66.6	54.3	43.0		
		MCWB	36.8	36.8	45.6	53.2	63.0	68.5	71.0	70.4	66.9	59.2	49.7	40.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	53.8	62.4	66.0	73.1	76.4	79.4	78.6	76.5	70.7	59.9	55.7		
		MCDB	54.2	57.4	70.5	78.2	84.4	88.4	89.2	87.2	85.6	78.5	63.0	57.4		
	2%	WB	48.5	47.1	56.2	61.8	70.2	74.2	76.7	76.0	73.2	67.0	56.7	50.3		
		MCDB	49.7	50.4	61.1	69.4	79.2	85.1	86.0	83.7	79.6	72.8	60.8	52.3		
	5%	WB	41.7	41.1	51.5	58.1	67.7	72.4	74.7	74.1	70.8	63.2	53.4	44.7		
		MCDB	43.8	44.2	56.3	66.0	76.5	82.3	83.3	81.0	77.0	69.0	57.0	47.3		
	10%	WB	37.0	37.0	46.0	54.4	64.9	70.6	73.1	72.4	68.7	60.2	50.0	40.3		
		MCDB	38.9	39.8	50.5	61.9	72.4	79.1	80.9	78.9	74.5	65.2	54.0	42.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	12.6	14.9	17.6	18.0	17.1	16.7	15.8	16.6	15.1	12.5	10.3		
		MCDBR	15.8	18.6	23.5	25.6	22.9	20.1	19.2	18.2	20.8	20.3	17.5	14.9		
		MCWBR	14.8	15.7	17.1	16.1	12.4	9.3	8.7	8.7	10.1	11.6	13.5	13.4		
	5% WB	MCDBR	15.6	17.7	22.5	23.1	20.7	18.3	17.3	15.9	17.5	18.1	16.1	14.5		
		MCWBR	15.2	15.9	17.6	16.2	12.8	9.8	9.1	8.9	10.3	11.4	13.6	14.0		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.335	0.353	0.399	0.433	0.448	0.438	0.430	0.389	0.360	0.320	0.312		
	taud		2.401	2.291	2.343	2.246	2.200	2.202	2.254	2.266	2.356	2.420	2.496	2.419		
	Ebn at Noon		266	272	282	276	268	264	265	263	267	261	259	251		
	Edh at Noon		25	33	35	42	45	45	43	41	35	29	23	23		
(44) All-Sky Solar Radiation	RadAvg		461	740	1109	1467	1718	1937	1935	1681	1356	882	548	385		
	RadStd		49	86	109	126	147	121	98	67	117	74	71	33		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+1.03	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+158	N/A

Nomenclature: See separate page